

Mohamed Amara

3B Mechatronics Engineering | Embedded Systems & Software | Winter 2027 Co-op

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Technical Skills

Languages: C, C++, Python, TypeScript, JavaScript, Java, MATLAB, Dart

Embedded & Robotics: STM32, ESP32, ARM Cortex-M, FreeRTOS, UART, SPI, I²C, CAN, PWM, ADC/DAC, sensors, PID

Software & Tools: React, Next.js, Node.js, Express, Flutter, MongoDB, Firebase, Supabase, Git, SolidWorks, KiCad, oscilloscopes, logic analyzers

Experience

AI & Hardware Systems Developer | The Plug Co

Sep. 2025–Dec. 2025, May–Aug. 2026

Python, TypeScript, C++, AI systems, embedded devices, system architecture

- Prototyped early VR hardware concepts using STM32 microcontrollers and breadboards to explore sensor integration, device I/O and hardware-software interaction.
- Evaluated custom hardware feasibility and helped shift development toward compatibility with existing Meta and commercial VR devices to reduce development time.
- Designed the end-to-end logic for capturing, synchronizing and structuring camera, audio, motion and user-interaction data for AI processing and searchable memories.
- Developed Python and TypeScript components to process device data, validate inputs and connect collected events with backend and AI-driven application features.

Software Engineering Intern | CoinWa

Jan. 2025–Apr. 2025

Flutter, Dart, Node.js, MongoDB, REST APIs

- Developed Flutter interfaces for interactive budgeting and saving modules across multiple mobile screen sizes.
- Implemented Node.js and MongoDB services for authentication, user accounts and learning-progress persistence.
- Improved application reliability by debugging rendering and performance issues with Flutter DevTools and end-to-end feature testing.

Software Engineering Intern | Western University

Feb. 2023–Jun. 2023

Java, requirements analysis, technical documentation

- Developed a Java allocation system that matched instructors to courses by qualifications and availability, improving scheduling efficiency by approximately 80%.
- Refined requirements with departmental staff, documented application workflows and trained users to support continued deployment.

Projects

OEM Wireless CarPlay Integration — 2016 Acura ILX

STM32, embedded systems, automotive electronics

- Designed an embedded architecture to add wireless CarPlay while retaining factory displays, controls and backup-camera functionality.
- Defined signal-measurement, video-switching and fail-safe requirements to enable factory-system fallback during hardware faults.

Armada — Real-Time Convoy Navigation Platform

Next.js, TypeScript, Node.js, Firebase

- Developed a mobile-first convoy platform with authentication, live locations, route visualization and synchronized driver profiles.
- Integrated geolocation, mapping and persistent cloud data to support real-time coordination across multiple connected users.

Human-Powered Washing Machine

SolidWorks, mechanical design

- Designed a pedal-driven drum and two-ratio chain drivetrain in SolidWorks, sizing shafts and transmission components using torque and fatigue analysis.

Education

University of Waterloo

Sep. 2023–Apr. 2028 Expected

Candidate for Bachelor of Applied Science in Honours Mechatronics Engineering

Relevant Coursework: Microprocessor Systems & Interfacing, Computer Structures & Real-Time Systems, Sensors & Instrumentation, Actuators & Power Electronics, Automatic Control Systems, Algorithms & Data Structures